

V075-E1-08

NS-Series Macro

PROGRAMMING MANUAL

OMRON

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Section1 Outline of Macro Function

This section describes execution conditions and programming procedure for using macro.

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1-1 What is Macro?

Macro is the function which can be executed by users original program. User can add functions, such as arithmetic operations and distinction of conditions, which are not supported by standard functions in CX-Designer. This function allows the PT to process screen display or data, which is performed by PLC before. It is also possible to reduce a load of PLC. In this manual, the timing for executing macro is called "Macro Execution Condition". Macro can be made by roughly divided three execution conditions as shown below.

- Execution condition for the project
- Execution condition for the screen
- Execution condition for the functional objects

There is no restriction on the number of macros for 1 project/1screen.
Up to 3000 characters can be used for one macro. Line feed is counted as two characters.
There is also no restriction on the number of lines.

Example:

'Number of inputting characters 11 characters+line

b feed (2characters), including comment
\$W0=10; 7characters+line feed (2characters)
STRCPY(\$W10",ABCDE"); 21characters
In this case, 43 characters are used.

1-2 Macro Execution Condition

Macros can be created for each project, screen, and functional objects. Also they can be created for the following execution conditions.

Executing conditions for the project

Macro execution conditions that can be made for the project are indicated below.

Select [PT]-[Project Properties]-[Macro] tab in CX-Designer, then set execution condition and record macro.

For details on registering macros, refer to the online CX-Designer Help 'System Settings and Project Properties'.

Execution condition	Explanation
When Loading a Project	Execute just before loading the first screen after starting up NS-Hardware
ON timing Alarm/Event occurred	Execute when alarm is occurred
ON timing Alarm/Event is canceled	Execute when alarm is cancelled
When a bit changed	Set macro to execute when the address of specified bit type is changed. Up to 10 macros can be set.
When a value changed	Set macro to execute when the address of specified word type is changed. Up to 10 macros can be set.

Execution conditions for the screen

Macro execution conditions that can be made for each screen are indicated below.

Select [PT]-[Screen/Sheet Properties]-[Macro] tab screen in CX-Designer, then set execution condition and record macro.

For details on registering macros, refer to the online CX-Designer Help 'Creating Screens'

Execution condition	Explanation
When Loading a Screen	Execute immediately after reading screen data to display the next
When Unloading a Screen	Execute immediately after closing the current screen

Reference

Macros are executed by the timing as shown below.

NS series, NSJ series and NSH series

	When loading a screen (Execute at the destination screen)	When unloading a screen (Execute at the destination screen)
User screen->User screen	Executed	Executed
User screen->Transfer screen	Not Executed	Executed
User screen->System menu	Not Executed	Executed
System menu->User screen	Executed	Not Executed
User screen->Screen Saver	Not Executed	Not Executed
Screen Saver->User screen	Not Executed	Not Executed

NS-Runtime

	When loading a screen (Execute at the destination screen)	When unloading a screen (Execute at the destination screen)
User screen->User screen	Executed	Executed
User screen->Transfer screen	Not Executed	Executed
User screen->System menu	Not Executed	Executed
System menu->User screen	Executed	Not Executed
User screen->Screen Saver	Not Executed	Not Executed
Screen Saver->User screen	Not Executed	Not Executed
When exiting NS-Runtime	-	Executed

Execution conditions for the functional objects

Macro execution conditions that can be made for each functional object are described in the following table.

Open property dialog for each functional object and select [Macro] tab page, then set the execution condition and create macro.

For details on registering macros, refer to the online CX-Designer Help 'Creating Functional Objects'.

Execution condition	Explanation
Touch on Timing	Execute when functional object is pressed.
Touch off Timing	Execute when functional object is released
Before Inputting numeral or character string	Execute just before display tenkey pad or virtual keyboard for inputting values or strings
Before Writing numeral or character string	Execute just before notice numeral and character string to the host.
When changing numeral and character string and comparing numeral	Execute when changing the value of address.
When Processing Display Area	Execute when display area for alarm display is pressed.
When Selecting an Alarm/Event	Execute just after select each alarm/event displayed on Alarm/Event Summary
When selecting a list	Execute just after select a list displayed on the List Selection.

Note

If the password is set for functional objects, the following macros are executed after inputting the password. If the password has been cancelled for inputting, macros will not be executed.

- Touch on/Touch off timing
- Before inputting Numeral/Character string
- When pressing Display Area
- When selecting an Alarm/Event
- When selecting a list

Section 1 Outline of Macro Function

NS series Macro Reference

The following conditions can be selected for functional objects.

Functional Object	Touch on Timing	Touch Off Timing	When changing Numeral/String and comparing Numeral	Before inputting Numeral/String	Before writing Numeral/String	List Selection
ON/OFF Button	OK	OK	-	-	-	-
Word Button	OK	OK	-	-	-	-
Command Button	OK	OK	-	-	-	-
Bit Lamp	-	-	OK	-	-	-
Word Lamp	-	-	OK	-	-	-
Numeral Display & Input	-	-	OK	OK	OK	-
String Display & Input	-	-	OK	OK	OK	-
Thumbwheel Switch	-	-	OK	-	OK	-
Text	-	-	-	-	-	-
List Selection	-	-	-	-	-	OK
Level Meter	-	-	-	-	-	-
Broken-line Graph	-	-	-	-	-	-
Bitmap	-	-	-	-	-	-
Analogue Meter	-	-	-	-	-	-
Video Display	-	-	-	-	-	-
Date	-	-	-	-	-	-
Time	-	-	-	-	-	-
Data Log Graph	-	-	-	-	-	-
Data Block Table	-	-	-	OK	OK	-
Temporary Input	-	-	-	-	-	-
Consecutive line drawing	-	-	-	-	-	-
Document Display	-	-	-	-	-	-
Multifunction Object	OK	OK	OK	-	-	-
Contents Display	-	-	-	-	-	-

[Alarm/Event object]

Functional Object	When Pressing a Display Area	When selecting an Alarm/Event
Alarm/Event Display	OK	-
Alarm/Event Summary	-	OK

Reference

- When a command button is set in the following functions, macros that are set at the touch on will be executed at the touch off. Macros that are set at the off will not be executed.

Switch screen
Control pop-up screen
Display system menu
Data block control

When functions other than above are selected, macros will be executed at the specified timing.

- Document Display is a functional object that can be used only with NS-Runtime.

1-3 Macro Programming

This section describes macro creation procedures and programming terms.

The Method for Writing a Macro

Delimiter of the Program

Put a semicolon (;) at the end of each program as a delimiter. However, it is not necessary for IF (), ELSEIF (), ELSE(),ENDIF.

Example;

\$W0=2;

IF (\$W0>=10)

\$W5=\$W0-\$W2;

ELSE

\$W5=\$W0+\$W2;

ENDIF

Comment

Put single quotation mark at the beginning of the sentence when you add the comment for each program. From single quotation mark (') to the end of character sting will be regarded as a comment.

Example:

\$W0 = 100; 'Comment

'Comment

IF (\$W1==200)

...

Writing Programming Terms

Both uppercase and lowercase can be used for programming macros because they are not classified.

Example:

-MovePopwDown () and MOVEPOPWDOWN() are regarded as same function.

-[Host1:DM0]and [host1:dm0] are regarded as same host address.

Programming Terms

This section describes terms used in this function.

Variable

The following variables can be used in macro program.

Item	Explanation
Host address	Use functions (READCMEM and WRITECMEM) for communication when accessing to the address in the host. Enclose address in [] Example: READCMEM(\$W100,[HOST1:DM00000],100); 'Read HOST1:DM00000 to DM00099 to \$W100 to \$W199
PT Memory	Bit Internal memory: \$B \$B0 to \$B32767 (1bit per 1point) Internal Holding memory: \$HB \$HB0 to \$HB8191 (1bit per 1point) System Memory: \$SB \$SB0 to \$SB 63 (1bit per 1point) Word Internal memory: \$W \$W to \$W32767(16bit per 1point) Internal Holding memory: \$HW \$HW0 to \$HW 8191 (16bit per 1point) System Memory: \$SW \$SW0 to \$SW39 (16bit per 1point) Example: \$W100=\$W0+1; 'Set the value \$W0 and 1 added to \$W100
Index	Index is used for processing bit and word in the PT memory. Add index to the end of the address and it will processed as [specified address + index value] There are 10 index points (I0 to I9). Set I0 to I9 for the value of \$SW27 to \$SW36. Example: \$SW27 = H20; \$W0I0 = 123; ' \$W0I0 is regarded as \$W20 added \$W0 and 20 '\$W20 = 123

Qualifier of Variable

Qualifiers set for variables must be used as shown below.

Qualifiers are used when performing 32-bit data processing and numeral processing for bit.

Item	Explanation
Long Access (32bit) of Word (16bit)	Put "L" at the end of variable. Uses 2 words. \$W0L=1000000; 'Accesses regarding \$W0, \$W1 as 32-bit \$W100L=1000*1000; 'Accesses regarding \$W100, \$W101 as 32-bit
Numeral Access of Bit	Put ":n" at the end of variable. Specify the value of bit address (up to 32 by 4-bit unit) for "n". Exceptions: If n=16, input "W". If n=32, input "L". \$B0:4 = 3; 'Sets 3(0011) for 4-bit from \$B0 to \$B3 \$B0W = 12345; 'Sets 12345(0011000000111001) for 16-bit from \$B0 to \$B15

Constant

Usable constants for macro program and procedure are described in the following table.

Item	Explanation
Decimal constant	-32768 to 32767 can be input when using word (16-bit) -2147483648 to 2147483647 can be input when using 2 words (32-bit)
Hexadecimal constant	H0 to HFFFF can be input when using word (16-bit) H0 to HFFFFFFFF can be input when using 2 words (32-bit)
Character String	Enclose in " " E.g. "ABCDE"

Branches

The following keywords can be used for specifying conditions.

Item	Explanation
IF ELSEIF ELSE ENDIF	Enclose conditional expressions in parentheses () after the IF and ELSEIF. Always use ENDIF at the end. Up to 8 nests can be input. There is no restrictions for inputting the number of lines under IF sentence. However, the total number of characters used in whole macro must be no more than 3000 characters. E.g. IF(\$W100 == 1) 'If \$W100 is 1 \$W99 = 1; ELSEIF(\$W100 == 2) 'if \$W100 is 2 \$W99 = 2; ELSE 'if \$W100 is other than 1 or 2 \$W99 = 3; ENDIF

Conditional Expressions

Use the following conditional expressions for specifying conditions in the IF sentences. It works for all types of data (word, long access of word, bit, and numeral access of bit).

Item	Explanation
A == B	If A is equal to B, TRUE.
A > B	If A is greater than B, TRUE.
A >= B	If A is greater than or equal to B, TRUE.
A < B	If A is less than B, TRUE.
A <= B	If A is less than or equal to B, TRUE.
A <> B A != B	If A is not equal to B, TRUE.
A && B A AND B	If both A and B are true, TRUE. (AND)
A B A OR B	If either of A or B is true, TRUE. (OR)

Reference

The result (A&&B, A>B) can be substituted for the variable.

E.g.\$B100=\$W0>100;

If the value of \$W is "100" or less, "0" will be substituted for \$B100. If the value of \$W is more than "100", "1" will be substituted for \$B100.

Basic Operational Statements

The following operational statements can be used in the program.

Item	Operator	Example	meaning
substitution	=	A = B	Substitute B for A
addition	+	C = A+B	Set A+B to C
subtraction	-	C = A-B	Set A-B to C
multiplication	*	C = A*B	Set AxB to C
division	/	C = A/B	Set A/B to C
residue	%	C = A%B	Set A%B to C
OR		C = A B	Logically ORs for A and B
AND	&	C = A & B	Logically ANDs for A and B
NOT	!	C = !A	Set C for denial of A
Exclusive	^	C = A^B	Result C of logical XORs for A and B
Complement of 1	~	B = ~A	Set the 1-complement of A to B
Bit Shift (left)	<<	C = A<<B	Set the value that A is ltic shifted B-bit to the left to C.
Bit Shift (right)	>>	C = A>>B	Se the value that A is arithmetic shifted B-bit to the right to C.

Reference

If executing logical operation, process must be performed between the same data types (between word, bit, or long access)

E.g. \$SW0L=\$SW10L&\$W20L;'Use all long access of word'

Multiple operations can be combined.

E.g. A=(B+C)*(D+E/2)

The priority of orders is as follows.

Item	Symbol
High	()
↑	~
	*, /, %
	+, -
	<<, >>
	&
	^
↓	
Low	=

Functions

The following functions are provided for macro of NS Series.

	Action	Function
Conversion between BCD and BIN	Value (BIN code)->BCD code	BCD
	BCD code->value (BIN code)	BIN
Manipulating character strings	Copy character string	STRCPY/STRCPYW
	Convert from ASCII code to Unicode	STRM2W
	Convert from Unicode to ASCII code	STRW2M
Alarm/Event summary	Clear the number of Alarm/Event occurrence	RSTALARMCNT
HMI exclusive statements	Output written value and changed value	GETNUMVAL
	Switch screen	SHOWPAGE/SHOWPAGEBCD
	Movement of object display area	MOVEPARTS
	Display message dialog box	MSGBOX
	Get displayed rectangle of the object	GETPARTS
	Move pop-up window	MOVEPOPW
	Move pop-up window up	MOVEPOPWUP
	Move pop-up window down	MOVEPOPWDOWN
	Move pop-up window left	MOVEPOPWLEFT
	Move pop-up window right	MOVEPOPWRIGHT
	Close pop-up window	CLOSEPOPW
Communications	Read data from specified address	READCMEM
	Write data to specified address	WRITECMEM
	Read bit data from specified address	READHOSTB
	Read word data from specified address	READHOSTW
	Write bit data to specified address	WRITEHOSTB
	Write word data to specified address	WRITEHOSTW
Process termination	Terminate macro program	RETURN
Set Date/Time	Change settings of internal clock of the PT	SETTIME
Reading/writing data	Read contents (values in binary) of the specified file in a memory card (CF) to PT memory.	READCF
	Save the contents of PT memory in a memory card (CF).	WRITECF
Write to multiple addresses	Write (0/1) to multiple bit addresses in the PT memory	BITSET
	Write a value to multiple word addresses in the PT memory	MEMSET

Section 1 Outline of Macro Function

NS series Macro Reference

Data manipulation /conversion	Swap high order and low order of the specified address.	SWAP
	Swap high order (2byte) and low order (2byte) of the specified long word data.	SWAPL
	Copy contents of \$W in the PT memory	MEMCOPY
Input Focus Control	Set the input focus for the specified object	SETFOCUS
	Release the input focus set for the object	RELEASEFOCUS
Repeat Program	Repeat Program	FOR, NEXT
	Aborting from Program Repetition	BREAK
	Return to the top of FOR loop.	CONTINUE

Section 1 Outline of Macro Function

NS series Macro Reference

The timing when a function can be executed as shown below.

	Project Macro			Screen Macro		Functional object Macro							
	When Loading a Project	Alarm/Event On Timing	Alarm/Event Off Timing	When Loading a screen	When Unloading a screen	Touch On Timing	Touch Off Timing	When changing value	Before Inputting Numeral/String	When writing Numeral/String	List Selection	When pressing a Display Area	When Selecting an Alarm/Event
BCD	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
BIN	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
CLOSEPOPW		OK	OK			OK	OK	OK	OK	OK	OK	OK	OK
GETNUMVAL								*		*			
GETPARTS				OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
MOVEPARTS				OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
MOVEPOPW		OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
MOVEPOPWDOWN		OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
MOVEPOPWLEFT		OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
MOVEPOPWRIGHT		OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
MOVEPOPWUP		OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
MSGBOX	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
READCMEM	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
RETURN	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
RSTALARMCNT	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
SHOWPAGE		OK	OK			OK	OK	OK	OK	OK	OK	OK	OK
SHOWPAGEBCD		OK	OK			OK	OK	OK	OK	OK	OK	OK	OK
STRCPY(W)	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
STRM2W	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
WRITECMEM	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
SETTIME	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
READCF	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
WRITECF	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
MEMCOPY	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
SWAP	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
SWAPL	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
SETFOCUS		OK	OK			OK	OK	OK			OK	OK	OK
RELEASEFOCUS		OK	OK			OK	OK	OK			OK	OK	OK
FOR, NEXT	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
BREAK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
CONTINUE	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
READHOSTB	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
READHOSTW	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
WRITEHOSTB	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
WRITE HOSTW	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
BITSET	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
MEMSET	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK

*Numeral Display & Input Only

Added Functions

Action	Function	Remarks
Set Date/Time	SETTIME	Added in the NS system program Ver.3.0
Reading/writing data	READCF	Added in the NS system program Ver.4.0
	WRITECF	
Data manipulation /conversion	SWAP	
	SWAPL	
	MEMCOPY	
Input Focus Control	SETFOCUS	Added in the NS system program Ver.5.0.
	RELEASEFOCUS	
Switch Screen (BCD)	SHOWPAGEBCD	Added in the NS system program Ver.6.0
Repeat Program	FOR, NEXT	
	BREAK	
	CONTINUE	
Communications	READHOSTB	Added in the NS system program Ver.6.2
	READHOSTW	
	WRITEHOSTB	
	WRITEHOSTW	
Write to multiple addresses	BITSET	
	MEMSET	

In addition to macro of NS Series, the following functions are provided for macro of NS-Runtime.

Action		Function
Manipulating character strings	String Comparison (Case sensitive)	STRCMP/STRCMPW
	String Comparison (Not case sensitive)	STRICMP/STRICMPW
	String Concatenation	STRCAT/STRCATW
	Gets String Length	STRLEN/STRLENW
	Extracts the specified number of characters from the leftmost characters of a string.	STRLEFT/STRLEFTW
	Extracts the specified number of characters from a specified character position of a string.	STRMID/STRMIDW
	Extracts the specified number of characters from the rightmost characters of a string.	STRRIGHT/STRRIGHTW
	Deletes the leftmost spaces of a string	STRLTRIM/STRLTRIMW
	Deletes the spaces at both sides of a string	STRTRIM/STRTRIMW
	Deletes the rightmost spaces of a string	STRRTRIM/STRRTRIMW
	Converts a string to lower case	STRLWR/STRLWRW
	Converts a string to upper case	STRUPR/STRUPRW
Manipulating Window	Finds a window title	WINFIND
	Maximizes a specified window	WINMAX
	Minimizes a specified window	WINMIN
	Restores a size of a specified window	WINNORMAL
	Brings a specified window to the front	WINTOP
	Exits a specified window	WINTERM
Module startup	Starts up an application	EXEC

Reference

These macros cannot be used with NS series, NSJ series and NSH series.

Section 1 Outline of Macro Function

NS series Macro Reference

The timing when a function can be executed as shown below.

	Project Macro			Screen Macro		Functional object Macro							
	When Loading a Project	Alarm/Event On Timing	Alarm/Event Off Timing	When Loading a screen	When Unloading a screen	Touch On Timing	Touch Off Timing	When changing value	Before Inputting Numeral/String	When writing Numeral/String	List Selection	When pressing a Display Area	When Selecting an Alarm/Event
STRCMP(W)	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
STRICMP(W)	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
STRCAT(W)	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
STRLEN(W)	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
STRLEFT(W)	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
STRMID(W)	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
STRRIGHT(W)	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
STRLTRIM(W)	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
STRTRIM(W)	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
STRRTRIM(W)	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
STRLWR(W)	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
STRUPR(W)	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
WINFIND	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
WINMAX	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
WINMIN	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
WINNORMAL	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
WINTOP	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
WINTERM	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK
EXEC	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK	OK

Section2 Functions

This section describes how to use the standard functions.

2-1	Table of Function and Argument.....	2
2-2	Details of the Function.....	7

2-1 Table of Function and Argument

The variety of variables and values, which can be specified as an argument for macro function is described below. Alphabets such as S,D,n,x,y in the row indicated parameter used in “2-2” Details of function”-“Format”.

Function	Argument	PT Memory				Constant	String	Host Side Address	Specify Index
		\$B \$HB \$SB	Numerical access of bit	\$W \$HW \$SW	Long access of word				
BCD	S		○	○	○	○			○
BIN	S		○	○	○	○			○
CLOSEPOPW	n		○	○	○	○			○
GETNUMVAL	None								
GETPARTS	N		○	○	○	○			○
	Left, Top, Right, Bottom			○	○				○
MOVEPARTS	n		○	○	○	○			○
	X		○	○	○	○			○
	Y		○	○	○	○			○
MOVEPOPW	n		○	○	○	○			○
	x		○	○	○	○			○
	y		○	○	○	○			○
MOVEPOPWDOWN	n		○	○	○	○			○
	y		○	○	○	○			○
MOVEPOPWLEFT	n		○	○	○	○			○
	x		○	○	○	○			○
MOVEPOPWRIGHT	n		○	○	○	○			○
	y		○	○	○	○			○
MOVEPOPWUP	n		○	○	○	○			○
	y		○	○	○	○			○
MSGBOX	S1			○	○		○		○
	S2			○	○		○		○
	S3		○	○	○	○			○

Function	Argument	PT Memory				Constant	String	Host Side Address	Specify Index
		\$B \$HB \$SB	Numerical access of bit	\$W \$HW \$SW	Long access				
READCMEM	D	○		○	○				○
	S	○		○	○			○	○
	N					○			
RETURN	S		○	○	○	○			○
RSTALARMCNT	S		○	○	○	○			○
SHOWPAGE	S		○	○	○	○			○
SHOWPAGEBCD	S		○	○	○	○			○
STRCPY(W)	D			○	○				○
	S			○	○		○		○
STRM2W	D			○	○				○
	S			○	○		○		○
STRM2M	D			○	○				○
	S			○	○		○		○
WRITECMEM	D	○		○	○			○	○
	S	○		○	○				○
	n	○	○	○	○				
SETTIME	S			○					
READCF	Mem			○					○
	Size			○					
	File			○			○		
	Dev			○		○			
WRITECF	Mem			○					○
	Size			○					
	File			○			○		
	Dev			○		○			
SWAP	S			○	○				○
	n			○	○	○			
SWAPL	S			○	○				○
	n			○	○	○			
MEMCOPY	S			○	○				○
	D			○	○				○
	N			○					○
SETFOCUS	N			○		○			○
RELEASEFOCUS	None								

Function	Argument	PT Memory				Constant	String	Host Side Address	Specify Index
		\$B \$HB \$SB	Numerical access of bit	\$W \$HW \$SW	Long access				
READHOSTB	D	○							○
	h			○		○		○	
	ch			○		○			○
	addr			○		○			○
	r			○		○			○
	n			○		○			○
READHOSTW	D			○					○
	h			○		○		○	
	ch			○		○			○
	Addr			○		○			○
	n			○		○			○
WRITEHOSTB	h			○		○		○	
	ch			○		○			○
	addr			○		○			○
	r			○		○			○
	S	○							○
	n			○		○			○
WRITEHOSTW	h			○		○		○	
	ch			○		○			○
	addr			○		○			○
	S			○					○
	n			○		○			○
BITSET	D	○							○
	c	○				○			○
	n			○		○			○
MEMSET	D			○					○
	c			○		○			○
	n			○		○			○

The following macros can be used only with NS-Runtime.

Function	Argument	PT Memory				Constant	String	Host Side Address	Specify Index
		\$B \$HB \$SB	Numerical access of bit	\$W \$HW \$SW	Long access of word				
STRCMP(W)	S1			○			○		○
	S2			○			○		○
STRICMP(W)	S1			○			○		○
	S2			○			○		○
STRCAT(W)	D			○					○
	S			○			○		○
STRLEN(W)	S			○			○		○
STRLEFT(W)	D			○					○
	S			○			○		○
	n		○	○	○	○			○
STRMID(W)	D			○					○
	S			○			○		○
	n1		○	○	○	○			○
	n2		○	○	○	○			○
STRRIGHT(W)	D			○					○
	S			○					○
	n		○	○	○	○			○
STRLTRIM(W)	D			○					○
	S			○			○		○
STRTRIM(W)	D			○					○
	S			○			○		○
STRRTRIM(W)	D			○					○
	S			○			○		○
STRLWR(W)	D			○					○
	S			○			○		○
STRUPR(W)	D			○					○
	S			○			○		○
WINFIND	S1			○			○		○
	S2		○	○	○	○			○
WINMAX	S1			○			○		○
	S2		○	○	○	○			○
WINMIN	S1			○			○		○
	S2		○	○	○	○			○

Function	Argument	PT Memory				Constant	String	Host Side Address	Specify Index
		\$B \$HB \$SB	Numerical access of bit	\$W \$HW \$SW	Long access of word				
WINNORMAL	S1			○			○		○
	S2		○	○	○	○			○
WINTERM	S1			○			○		○
	S2		○	○	○	○			○
WINTOP	S1			○		○		○	
	S2			○	○			○	
Exec	S1			○			○		○
	S2			○			○		○
	S3		○	○	○	○			○

2-2 Details of the Function

This section describes details of standard functions used in Macro programming.

BCD Converts the Value to BCD code

Applicable versions	System Version 2 or higher
Format	BCD(S)
Function	Convert value "s" to BCD code Converting range is 0 to 99999999 If you specify the character string outside of the range, overflow occurs. "0" is set at the end of the character string.
Return Value	BCD code
Example	<pre>\$W0 = 1234; 'Set value 1234 to \$W0 \$W10 = BCD(\$W0); 'Set BCD code (H1234) to \$W10 \$W20L = 12345678; 'Set value 12345678 to \$W20 to \$W21 \$W22L = BCD(\$W20L); 'Set BCD code (H12345678) to \$W22 to W23</pre>

BIN Converts BCD code to Numeral value

Applicable versions	System Version 2 or higher
Format	BIN(S)
Function	Convert BCD code S to numeral value Converting range is H0 to H99999999
Return Value	Numeral value
Example	<pre>\$W0 = H1234; 'Set BCD code (H1234) to \$W0 \$W10 = BIN(\$W0); 'Set 1234 to \$W10 \$W20L = H334455; 'Set BCD code (H334455) to \$W20 to \$W21BCD \$W22L = BIN(\$W20L); 'Set 334455 to \$W22 to \$W23</pre>

BITSET **Writes (0/1) to multiple bit addresses in the PT memory**

Applicable versions	System Version 6.2 or higher
Format	BITSET(D, c, n)
Function	Writes c(0/1) for n-bit data from bit address D in the PT memory (\$B/\$HB). D: Starting address c: Set value (0/1) n: Number of elements to write 1 to 32768 (\$B) 1 to 8192 (\$HB)
Return Value	None
Example	- Writing 1 to 10 bits from \$B100 (\$B100 to \$B109) BITSET(\$B100, 1, 10); - Writing 1 to 10 bits from \$B100 (\$B100 to \$B109) \$HB100=1; \$W200=10; BITSET(\$B100, \$HB100, \$W200);

BREAK **Aborts from program repetition**

Applicable versions	System version 6 or higher
Format	BREAK
Function	Interrupt a loop program between “FOR and NEXT”.
Return Value	None
Example	If \$W100I0>30 is true, exit FOR loop. <pre> \$SW27=0; FOR(10) \$W100I0=\$W50I0+10; IF(\$W100I0>30) BREAK; ENDIF \$SW27=\$SW27+1; NEXT; </pre> *Setting range for “n” is 0 to 32767. A negative number is considered as 0. \$W, \$HW and \$SW can specified as an address.

CLOSEPOPW **Closes pop-up window**

Applicable versions	System version 2 or higher
Format	CLOSEPOPW(n)
Function	Close pop-up window screen page number “n” Setting range for “n” is 0 to 3999. If you set pop-up screen page number that does not exist, the process will not be performed.
Return Value	None
Example	CLOSEPOPW(15);‘Close pop-up screen page 15

CONTINUE **Repeats program**

Applicable versions	System version 6 or higher
Format	CONTINUE
Function	During a program between “FOR to NEXT”, it will return to the top of the FOR loop and resume the FOR process.
Return Value	None
Example	If '\$W50I0>30 is true, a loop will return to the top, and resume the next repetitious program. <pre> \$SW27=0; FOR(10) IF(\$W50I0>30) \$SW27=\$SW27+1; CONTINUE; ENDIF \$W100I0=\$W50I0+10; \$SW27=\$SW27+1; NEXT; </pre> *Setting range for “n” is 0 to 32767. A negative number is considered as 0. \$W, \$HW and \$SW can be specified as an address.

FOR(n), NEXT n; number of iteration

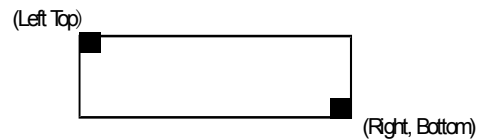
Applicable versions	System version 6 or higher
Format	FOR(n), NEXT n; a LOOP
Function	A series of statements in a computer program that are to be executed repeatedly at specified times. A program between “FOR to NEXT” cannot be nested in another “FOR to NEXT”. (Single loop only)
Return Value	None
Example	Execute a loop ”FOR to NEXT” 10 times and substitute \$W0~’\$W9 to the initial value,0. <pre> \$W0=0; \$SW27=0; FOR(10) \$W0I0=0; \$SW27=\$SW27+1; NEXT; </pre> *Setting range for “n” is 0 to 32767. A negative number is considered as 0. \$W, \$HW or \$SW can be specified as an address.

GETNUMVAL Outputs written value and changed value

Applicable versions	System version 2 or higher
Format	GETNUMVAL()
Function	Get writing numeral value or changing numeral value for numeral display & input object. Use this function for “Before writing numeral” or “When changing numeral” at “Macro Execution Condition” in numeral display & input object.
Return Value	Input numeral value
Example	<pre> \$W0=GETNUMVAL(); </pre> ‘Set value for writing numeral value to \$W0

GETPARTS Gets displayed rectangle of the object

Applicable versions	System version 2 or higher
Format	GETPARTS(n, Left, Top, Right, Bottom)
Function	Get displayed rectangle of the object ID number “n”. Set coordinate (Left, Top) at the upper left, (Right, Bottom) at the lower right on rectangle.



Setting range is 0 to 1023. If you set other value or ID number that does not exist, return value 1 will be returned.

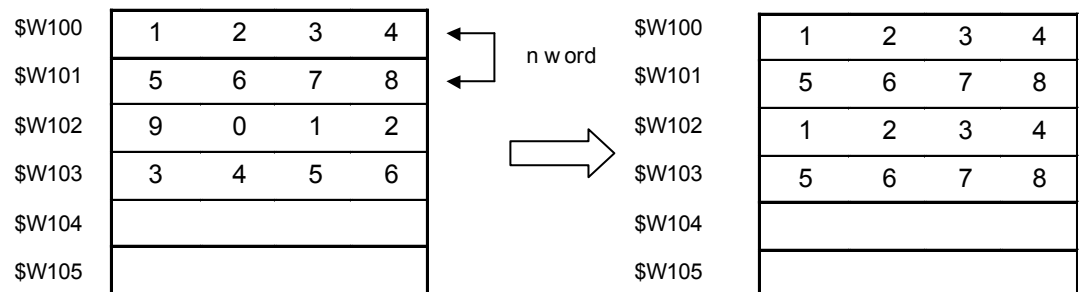
Return Value	0: Completed normally -1:Specified no object
Example	GETPARTS(1, \$W0, \$W1, \$W2, \$W3); 'Set coordinate of displayed rectangle of object ID number1 'to (\$W0, \$W1)-(\$W2, \$W3)

MEMCOPY **Copies contents of \$W in the PT memory**

Applicable versions	System version 4 or higher
Format	MEMCOPY (S, D, n);
Function	Copy data of \$W or \$HW in the PT memory. S: Top address of source data. D: Top address to which data will be copied. n: The number of word data which will be taken from S. Setting range is as follows: When specifying "n" directly: 1 to 32767 When specifying "n" indirectly: \$W0 to \$W32767 \$HW0 to \$HW8191 Note: Index can be set when using address to specify for S. Setting range is for \$W is between 0 and 32767 and for \$HW is between 0 to 8191.
Return Value	None
Example	MEMCOPY(\$W100, \$W102, 2); Take 2 words from \$W 100 and copy to 102

<Before executing MEMCOPY>

<After executing MEMCOPY>



MEMSET Writes a value to multiple word addresses in the PT memory

Applicable versions	System Version 6.2 or higher
Format	MEMSET(D, c, n)
Function	Writes data c for n-word from a word address D in the PT memory (\$W/\$HW). D: Starting address c: Set value -32767 to 32768 (decimal format) H0000 to HFFFF (hexadecimal format) n: Number of elements to write 1 to 32768 (\$W) 1 to 8192 (\$HW)
Return Value	None
Example	- Writing 5 for 10 words from \$W100 (\$W100 to \$W109) MEMSET(\$W100, 5, 10); - Writes 5 for 10 words from \$W100 (\$W100 to \$W109) \$HW100=5; \$W200=10; MEMSET(\$W100, \$HW100, \$W200);

MOVEPARTS Moves object display area

Applicable versions	System version 2 or higher
Format	MOVEPARTS (n,x,y)
Function	Move the object ID number “n” to specified coordinate (x, y). Specify coordinate upper left of the moving object for “x, y”. Setting range for “n” is 0 to 1023. If the value out side the range or ID number that does not exist is specified, return value –1 will be returned. There is no restriction on setting value for “x, y”. However, set the value for the coordinate of x and y in order that the objects are displayed inside of the screen. All objects or some objects on the screen may be deleted depending on the set value so care must be taken.
Return Value	0: Completed normally -1: Specified no object
Example	MOVEPARTS (3, 150, 200); Move the object ID number 3 to position (150,200)

Reference

When setting macro “MOVEPARTS” for ON/OFF button, Word button, Command button and Multifunction Object and moving these objects or these objects with frame, select “Touch Off Timing”. If “Touch On Timing” is selected, the status of the object will be pressed.

MOVEPOPW **Moves pop-up window**

Applicable versions	System version 2 or higher
Format	MOVEPOPW(n,x,y)
Function	Moves top left of the pop-up window for screen number “n” to the specified coordinates (x, y). Setting range for “n” is 0 to 3999. If the value outside of the range or screen number that does not exist is specified, return value “-1” will be returned. There is no restriction on setting the value for “x, y”. However, set the value for the coordinate of x and y in order that the objects are displayed inside of the screen. Part of the screen or whole screen may be deleted depending on the set value.
Return Value	0: Completed normally -1: Specified no page
Example	\$W0=MOVEPOPW(10, 140, 160); ‘Moving pop-up screen page number 10 to the specified position (140, 160), then return “0” to \$W0. ‘If pop-up screen is not displayed, return “-1” to \$W0.

Reference

When setting macro “MOVEPOPW” for ON/OFF button, Word button, Command button and Multifunction Object on a pop-up screen and moving the pop-up screen, select “Touch Off Timing”. If “Touch On Timing” is selected, the status of the object will be pressed.

MOVEPOPWDOWN Moves pop-up window down

Applicable versions	System version 2 or higher
Format	MOVEPOPWDOWN(n, y)
Function	Move the pop-up window page number “n” to y down. Setting range for “n” is 0 to 3999. If the value outside of the range or page number which does not exist is specified, return value “-1” will be returned. There is no restriction on setting value for “y”. However, set the value to the pop-up screen in order to be displayed inside of the screen. Part of the screen or whole screen may be deleted depending on the set value.
Return Value	0: Completed normally -1: Specified no page
Example	\$W0=MOVEPOPWDOWN(10, 32); ‘Move the pop-up window page number 10 to 32dot down, ‘then return “0”to \$W0. If pop-up screen is not displayed, ‘return “-1” to #W0.

Reference

When setting macro “MOVEPOPWDOWN” for ON/OFF button, Word button, Command button and Multifunction Object on a pop-up screen and moving the pop-up screen, select “Touch Off Timing”. If “Touch On Timing” is selected, the status of the object will be pressed.

MOVEPOPWLEFT Moves pop-up window to the left

Applicable versions	System version 2 or higher
Format	MOVEPOPWLEFT (n, x)
Function	Move the pop-up window page number “n” to x dot left. Setting rage for “x” is 0 to 3999. If the value outside of the range or page number that does not exist is specified, return value “-1” is returned. There is no restriction on setting the value for “x”. However, set the value in order to be displayed inside of the screen. It may be deleted part of the screen or whole screen depends on the value.
Return Value	0: Completed normally -1: Specified no page
Example	\$W0=MOVEPOPWLEFT (10, 32); 'Move pop-up window page number 10 to 32 dot left, then return "0" to \$W0. If pop-up screen is not displayed, return "-1" to \$W0.

Reference

When setting macro “MOVEPOPWLEFT” for ON/OFF button, Word button, Command button and Multifunction Object on a pop-up screen and moving the pop-up screen, select “Touch Off Timing”. If “Touch On Timing” is selected, the status of the object will be pressed.

MOVEPOPWRIGHT Moves pop-up window to the right

Applicable versions	System version 2 or higher
Format	MOVEPOPWRIGHT(n, x)
Function	Move pop-up window page “n” to x dot right. Setting range for “n” is 0 to 3999. If the value outside the range or screen number that does not exist is specified, return value “-1” is returned. There is no restriction on setting for “x”. However, set the value in order to be displayed inside of the screen. Part of the screen or whole screen may be deleted depending on the set value.
Return Value	0: Completed normally -1: Specified no page
Example	\$W0=MOVEPOPWRIGHT (10,32); ‘Move the pop-up screen page 10 to 32 dot right, then return “0” to \$W0. If the pop-up screen is not displayed, return “-1” to \$W0.

Reference

When setting macro “MOVEPOPWRIGHT” for ON/OFF button, Word button, Command button and Multifunction Object on a pop-up screen and moving the pop-up screen, select “Touch Off Timing”. If “Touch On Timing” is selected, the status of the object will be pressed.

MOVEPOPWUP Moves pop-up window up

Applicable versions	System version 2 or higher
Format	MOVEPOPWUP (n, y)
Function	Move the pop-up screen page “n” to y dot up. Setting range for “n” is 0 to 3999. If the value outside of the range or page number that does not exist is specified, return value “-1” is returned. There is no restriction on setting the value for “y”. However, set the value in order to be displayed inside of the screen. Part of the screen or whole screen may be deleted depending on the set value.
Return Value	0: Completed normally -1: Specified no page
Example	\$W0=MOVEPOPWUP (10,32); ‘Move pop-up screen page 10 to 32 dot up, then return “0” to \$W0. If pop-up screen is not displayed, return “-1” to \$W0.

Reference

When setting macro “MOVEPOPWUP” for ON/OFF button, Word button, Command button and Multifunction Object on a pop-up screen and moving the pop-up screen, select “Touch Off Timing”. If “Touch On Timing” is selected, the status of the object will be pressed.

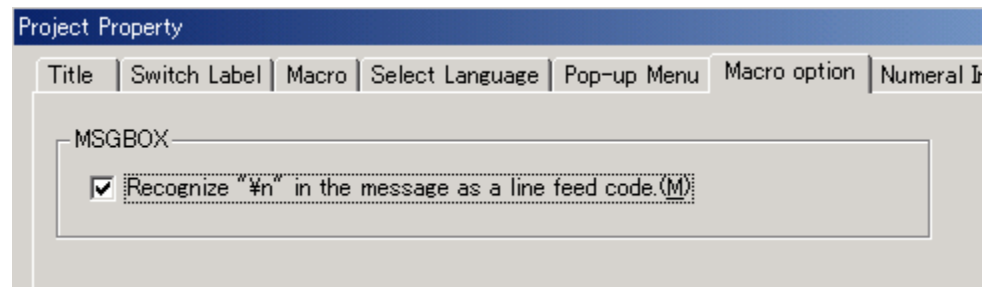
MSGBOX Displays message dialog box

Applicable versions	System version 2 or higher
Format	MSGBOX (S1, S2, S3)
Function	Display message dialog which is specified.

S1: message string

Perform the following procedure to insert a line feed in the message.

Select *PT – Project Properties* in the CX-Designer to display the Project Property Dialog box. Checking “Recognize “\n” in the message as a line feed code” of the MSGBOX option in the Macro option tab enables to insert a line feed by typing “\n” in the message string.



S2: title bar string

S3: reply with icon type that is displayed in message dialog

Specify type of the button.

4 bits (B0-B3)

0:	STOP Mark	1:	QUESTION Mark
2:	EXCLAMATION Mark	3:	INFORMATION Mark

4 bits (B4-B7)

0:[OK] button only

1:[OK]/[Cancel]button

2:[Retry]/[Cancel]button

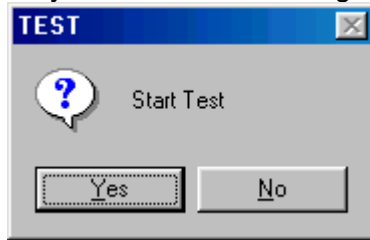
3:[Yes]/[No]button

4:[Yes]/[No]/[Cancel]button

5:[Stop]/[Retry]/[Ignore]button

Return Value	0:Select [OK] button	1:Select [Cancel]button
	2:Select [Yes]button	3:Select [No]button
	4:Select [Ignore]button	5:Select [Retry]button
	6:Select [Stop]button	

Example	<pre> \$W0=MSGBOX ("Start Test", "TEST", H31); 'H31:[Yes]/[No] button, Display QUESTION mark IF(\$W0==2) 'Write processing if you select "Yes" ELSE 'Write processing if you select "No" ENDIF As you set above, message dialog is displayed below </pre>
---------	---



Reference

Only one message box can be displayed using MSGBOX. If executing MSGBOX with displaying other message box, new message box is not displayed and "1" is returned as return value.

Example: making two bit lamps

	address	macro
Lamp 1	\$B0	\$W0=MSGBOX("message1", "title1", H31);
Lamp 2	\$B0	\$W1=MSGBOX("message2", "title2", H31);

Suppose macro of lamp1 is executed first. When changing the value of \$B0, the message box of lamp 1 is displayed. Message box of lamp 2 is not displayed and value "1" is stored in \$W1.

If Buzzer Sound is set ON or OFF at PT tab in the system menu and STOP or EXCLAMATION is specified for the icon, buzzer will be sounded when displaying the message dialog.

READCF Reads the contents (values in binary) of the specified file in a memory card (CF) to PT memory

Applicable versions	System version 4 or higher
Format	READCF(Mem, Size, File, Dev)
Function	Reads the contents (binary format) of the specified file in a Memory Card to PT memory. Mem: Top address of destination. (\$W, \$HW or \$SW) Index can be set. Setting range for \$W is between 0 and 32767 and for \$HW is between 0 to 8191. \$B, \$HB and \$SB cannot be specified. Size: Data size to be read. (unit: word) Data size can be specified using long word directly, or \$W or \$HW(uses 2 words) indirectly. Setting range for \$W is 0 to 32767 and for \$HW is between 0 and 8191. If the set value is exceeded the maximum, an access error occurs and an error message appears. If the value 0 or less is set for Size, the specified size will be read to the PT memory. If the file size is bigger than the value set for Size (Size>0), it executes reading data set for Size. If the file size is smaller than the value set for Size(Size>0), it executes reading actual file size only. File: Source file name The file name can be specified using character string directly, or \$W or \$HW (uses 2 words) indirectly. Read action is executed by word unit, however, the last 1 byte of address will not be read if file size is odd byte. (Return value will be 0). Up to 43 alphanumerical characters ("0 to 9", "A to Z", "a to z", "\$", "_") including extension can be set for the file name. Dev: Specify destination device. Always specify 0 since destination will be a memory card only.
Return Value	0: Completed normally -1: Failed to read data
Example	\$W100=READCF(\$W1000,0", CF_FILE.BIN",0); \$W2000L=0; STRCOPY(\$W2002, "CF_FILE.BIN"); \$W100=READCF(\$W1000,\$W2000,\$W2002,0);

READCMEM Reads the data from specified address

Applicable versions	System version 2 or higher
Format	READCMEM (D,S,n)
Function	Read n but/n channel from the address in the host specified with “s”, and copy to “D”. Maximum points for reading are indicated below.

Bit	126Bit
Word	126Channel

If value outside of the range is set for “n”, communication error or macro execution error occurs.
Maximum points of reading depend on PLC type.

Return Value	None
Example	READCMEM (\$W0, [HOST1:DM0], 10) ‘Read the value “\$W0 to \$W9” to “DM0 to DM9” at the host ‘named Host1 in PLC.

READHOSTB Reads bit data from the specified address

Applicable versions	System Version 6.2 or higher
Format	READHOSTB(D, h, ch, addr, r, n)
Function	Reads n-bit data from the host, h and copies it to the PT memory (\$B/\$HB), D. D: Starting address to read data to (\$B0 to \$B32767, \$HB0 to \$HB8191) h: Host (host name / host number) ch: Host address type *1 addr: Starting address in the host r: Bits n: Number of elements to write (1 to 126) *1: Refer to <i>Address Type Number</i> at the end of this chapter.
Return Value	Normal termination: 0x0000 Error: high order 8 bits (B8 to B15): MRES (main response cord) low order 8 bits (B0 to B7) : SRES (sub-response cord) *Refer to <i>5-2-7 Communications Errors and Countermeasures</i> in the <i>NS-Series Programming Manual</i> for MRES and SRES.
Example	SerialA and Serial B are registered in the host: - Reads 10-bit data from CIO1000.00 in the PLC connected to the host 1 (Serial port A) and stores it to \$B10 to \$B19. READHOSTB(\$B10, 1, 100, 1000, 0, 10); - Reads 10-bit data from DM2000.05 in the PLC connected to the host name=[Serial B] (Serial port B) and stores it to \$HB10 to \$HB19. READHOSTB(\$HB10, [SerialB], 300, 2000, 5, 10);

Reference

- ◆ If a host is deleted with CX-Designer, the subsequent hosts will be renumbered in order. The example below shows when HOST2 is deleted, the host numbers are renumbered and HOST3's number will be changed to 2.

Before Deleting Host		After Deleting Host	
Host Number	Host Name	Host Number	Host Name
1	SERIALA	1	SERIALA
2	HOST2	2	HOST3
3	HOST3	-	-

Please note that when a host number is given as argument h of READHOSTB, it is necessary to specify the host number after change. With the above example, please change the macro as follows.

Macro Before Deleting Host	READHOSTB(\$B10,3,100,1000,0,10);	Change "3", the 2 nd argument that specifies HOST3, to "2"
Macro After Deleting Host	READHOSTB(\$B10,2,100,1000,0,10);	

To check the host number, confirm the number at the left side of host name shown on the Comm.Setting dialog of CX-Designer.

- ◆ Also, note that if a host name is changed with CX-Designer when the host name is used as argument h of READHOSTB, it is necessary to specify the host name after change.

For example, if a host name is changed from HOST_1 to HOST_A, please change as follows.

Macro Before Changing Host Name	READHOSTB(\$B10,[HOST_1],100,1000,0,10);	Change HOST_1 to HOST_A
Macro After Changing Host Name	READHOSTB(\$B10,[HOST_A],100,1000,0,10);	

READHOSTW Reads word data from the specified address

Applicable versions	System Version 6.2 or higher
Format	READHOSTW(D, h, ch, addr, n)
Function	Reads n-word data from the host (h) and copies it to the PT memory (\$W/\$HW), D. D: Starting address to read data to (\$W0 to \$W32767, \$HW0 to \$HW8191) h: Host (host name/host number) ch: Host address type *1 addr: Host starting address n: Number of elements to write (1 to 126) *1: Refer to <i>Address Type Number</i> at the end of this chapter.
Return Value	Normal termination: 0x0000 Error: high order 8 bits (B8 to B15): MRES (main response cord) low order 8 bits (B0 to B7) : SRES (sub-response cord) *Refer to <i>5-2-7 Communications Errors and Countermeasures</i> in the <i>NS-Series Programming Manual</i> for MRES and SRES.
Example	SerialA and Serial B are registered in the host: - Reads 10-bit data from CIO1000 in the PLC connected to the host 1 (Serial port A) and stores it to \$W10 to \$W19. READHOSTW(\$W10, 1, 100, 1000, 10); - Reads 10-bit data from DM2000 in the PLC connected to the host name=[Serial B] (Serial port B) and stores it to \$HW10 to \$HW19. READHOSTW(\$HW10, [SerialB], 300, 2000, 10);

Reference

- ◆ If a host is deleted with CX-Designer, the subsequent hosts will be renumbered in order. The example below shows when HOST2 is deleted, the host numbers are renumbered and HOST3's number will be changed to 2.

Before Deleting Host		After Deleting Host	
Host Number	Host Name	Host Number	Host Name
1	SERIALA	1	SERIALA
2	HOST2	2	HOST3
3	HOST3	-	-

Please note that when a host number is given as argument h of READHOSTW, it is necessary to specify the host number after change. With the above example, please change the macro as follows.

Macro Before Deleting Host	READHOSTW(\$B10, <u>3</u> ,100,1000,0,10);	Change "3", the 2 nd argument that specifies HOST3, to "2"
Macro After Deleting Host	READHOSTW(\$B10, <u>2</u> ,100,1000,0,10);	

To check the host number, confirm the number at the left side of host name shown on the Comm.Setting dialog of CX-Designer.

- ◆ Also, note that if a host name is changed with CX-Designer when the host name is used as argument h of READHOSTW, it is necessary to specify the host name after change.

For example, if a host name is changed from HOST_1 to HOST_A, please change as follows.

Macro Before Changing Host Name	READHOSTW(\$B10,[HOST_1],100,1000,0,10);	Change HOST_1 to HOST_A
Macro After Changing Host Name	READHOSTW(\$B10,[HOST_A],100,1000,0,10);	

RELEASEFOCUS Releases the input focus set for the object

Applicable versions	System version 5 or higher
Format	RELEASEFOCUS(); No argument is used.
Function	If the input focus has been set for any of numeral display & input object or string display & input object in the project, this macro will release the input focus. - If an object that the input focus has been set exists on the screen currently displayed, the macro will release the input focus. - If the input focus is not set for any object on the screen currently displayed, this macro will NOT work. - If the input focus has been set for the object created in the frame page displayed as top, this macro will also release the focus. - If the input focus has been set for the object in the sheet, this macro will release the focus. RELEASEFOCUS macro will NOT work in the following execution timing. -When Loading a Project -When Loading a Screen -When Unloading a Screen -Before Inputting Numeral set using numeral display & input objects -Before Writing Numeral set using numeral display & input objects -Before Inputting String set using string display & input objects -Before Writing String set using string display & input objects
Return Value	None
Example	In all cases that you want to release the input focus, set as the following example. RELEASEFOCUS();

RETURN Terminates Macro program

Applicable versions	System version 2 or higher
Format	RETURN(S)
Function	If the value of "S" is "0", terminate macro program and continue to process for functional object. If a value is set other than "0", terminate program and stop processing for functional object.
Return Value	None
Example	RETURN(0); 'terminate macro and continue to process RETURN(1); 'terminate macro and stop processing

RSTALARMCNT Clears the number of occurrence of Alarm/Event

Applicable versions	System version 2 or higher
Format	RSTALARMCNT(S)
Function	When the value of "S" is 0, clear the number of occurrence of alarm. When the value of "s" is 1, clear the number of occurrence of event.
Return Value	0: Completed normally -1: None
Example	RSTALARMCNT(0); 'clear the number of occurrence of alarm RSTALARMCNT(1); 'clear the number of occurrence of event

SETFOCUS Sets the input focus set for the object

Applicable versions	System version 5 or higher
Format	SETFOCUS(n);
Function	Set the input focus on the specified numeral display & input object or string display & input object. n: Object ID number which the input focus should be set. (0 to 32767) When setting the input focus for the object specified as top in the Input Order List, set "-1". - This macro will NOT work if the input focus has already been set for other object. - The input focus will NOT be set if the specified object is created in the frame page which is not displayed as top. (A dialog which indicates macro execution error will be displayed when executing this macro.) - If an object other than numeral display & input object and string display & input is specified, this macro will NOT work. - The input focus cannot be set for the objects created in the sheet. SETFOCUS macro will NOT work in the following execution timing. -When Loading a Project -When Loading a Screen -When Unloading a Screen -Before inputting Numeral set using numeral display & input objects -Before Writing Numeral set using numeral display & input objects -Before inputting String set using string display & input objects -Before Writing String set using string display & input objects
Return Value	0: Completed normally -1: The specified object ID could not be found.
Example	Case that the input focus is set for object with ID number 4. SETFOCUS(4);

SETTIME Changes settings of internal clock of the PT

Applicable versions	System version 3 or higher
Format	SETTIME(S)
Function	Preset values for the specified address as S and writes them to the internal clock of PT. Setting range for S is between \$W0 and \$W32765 or between \$HW0 and \$HW8189. (See note) Set the value in BCD format for addresses to be written. Uses 3 words regarding the specified address as top. Note: Index can be used for specifying the address.
Return Value	None
Example	Case that December 31, 2002 18:59:20 is set. \$W 100=H5920; \$W101=H3118; \$W102=H0212; SETTIME(\$W100);

SHOWPAGE Switches screen

Applicable versions	System version 2 or higher
Format	SHOWPAGE(S)
Function	Switch screen to the page that is specified in “s”. Setting range for “s” is 0 to 3999. If the value outside of the range is set, macro execution error occurs. If the screen number that does not exist is set, reading page error occurs.
Return Value	None
Example	SHOWPAGE(10); ‘Switch screen to page 10

Reference

Macro written after SHOWPAGE is not executed. Be sure to write SHOWPAGE at the end of line.

Bad Example:

```
SHOWPAGE(10); <-Switch to page 10
$W50=100;      <-Substitute 100 to $W50 is not executed
```

Good Example:

```
$W50=100;      <-Substitute 100 to $W50 is executed
SHOWPAGE(10); <-Switch to page 10
```

Similarly, when SHOWPAGE is executed by a macro set with Multifunction object, functions set after this macro will not be executed. In order to switch screens using SHOWPAGE, set this macro as the last function to be executed.

SHOWPAGEBCD Switches screen to the screen page n.

Applicable versions	System version 6 or higher
Format	SHOWPAGEBCD(S)
Function	S: screen page number (H0 to H3999) Switch screen to the page that is specified in "S". Setting range for "S" is H0 to H3999. If either the value outside of the range is set or an invalid value is set for BCD, macro execution error occurs. If the screen number that does not exist is set, reading page error occurs.
Return Value	None
Example	Switch screen to page 10 SHOWPAGEBCD(H10); Specify a screen page number indirectly to switch screen to page 10. \$W100=H10; SHOWPAGEBCD(\$W100);

Reference

Macro written after SHOWPAGEBCD is not executed. Be sure to write SHOWPAGEBCD at the end of line.

Bad Example:

SHOWPAGEBCD(H10); ←Switch to page 10
\$W50=100; ←Substitute 100 to \$W50 is not executed

Good Example:

\$W50=100; ←Substitute 100 to \$W50 is executed
SHOWPAGEBCD(H10); ←Switch to page 10

Similarly, when SHOWPAGEBCD is executed by a macro set with Multifunction object, function set after this macro will not be executed. In order to switch screens using SHOWPAGEBCD, this macro as the last function to be executed.

STRCPY(W) Copies Character string

Applicable versions	System version 2 or higher
Format	STRCPY (D,S) ;ASCII code STRCPYW(D,S) ;Uni code
Function	Copy character string from D to S Copy is performed including null.
Return Value	None
Example	STRCPYW (\$W0",ABC"); 'Set "ABC" to \$W0 to \$W2 \$W100=H6400;STRCPY(\$W110,\$W100); 'Set "d" to \$W110

Reference

'null' matches for "00" in ASCII code, and "0000" in Unicode.
Care must be taken to set string to \$W32767 because copy is performed including null.
When executing STRCPY(W), string data and null may not be executed \$W32767
because null is copied. (If data is exceeded \$W32767, communication error occurs).

STRM2W Converts character string from ASCII code to Unicode

Applicable versions	System version 2 or higher
Format	STRM2W (D,S)
Function	Convert character string specified in "S" from ASCII code to Unicode and copy to "D". Copy is performed including null.
Return Value	None
Example	STRM2W(\$W0", ABC"); 'Convert "ABC" to Unicode, and copy to \$W0 to \$W2

STRW2M Converts character string from Unicode to ASCII code

Applicable versions	System version 2 or higher
Format	STRW2M (D,S)
Function	Convert string specified in “S” to ASCII code and copy to “D”. Copy is performed including null.
Return Value	None
Example	STRW2M(\$W0, “ABC”); ‘Convert “ABC “ to ASCII code and copy to \$W0 to \$W1.

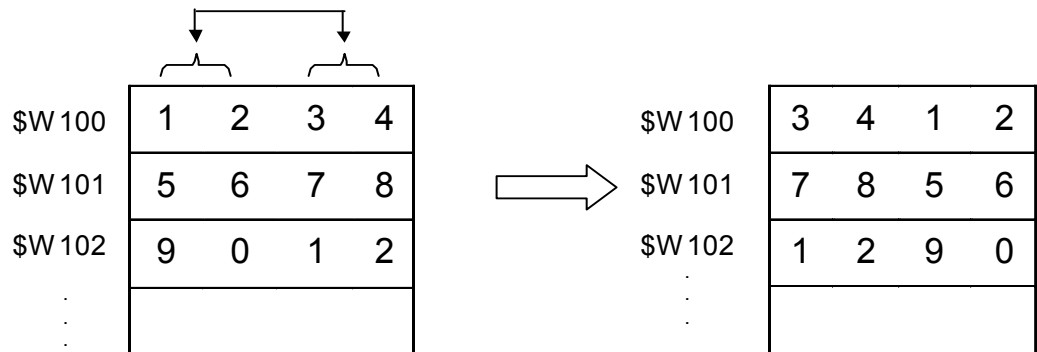
SWAP Swaps high order and low order of the specified address

Applicable versions	System version 4 or higher
Format	SWAP(S,n)
Function	Swap high order (1 byte) and low order (1 byte) of the word data or the internal holding word which was taken n word from S. S: Top address (\$W or \$HW) to be swapped. (See note.) n: The number of words to be taken from S. Setting range is as follows: When specifying "n" directly: 1 to 32767 When specifying "n" indirectly: \$W0 to \$W32767 \$HW0 to \$HW8191 Note: Index can be set when using address to specify for S. Setting range is for \$W is between 0 and 32767 and for \$HW is between 0 to 8191.
Return Value	None
Example	SWAP(\$W100,3); Swap high order and low order of the word data which was taken from 3 words from \$W100.

<Before performing SWAP>

<After performing SWAP>

Swap high order and low order



SWAPL Swaps high order (2byte) and low order (2byte) of the specified long word data

Applicable versions	System version 4 or higher
Format	SWAPL(S,n)
Function	Swap high order (2 byte) and low order (2 byte) of the long word data or the internal holding word which was taken n long word from S. S: Top address (\$W or \$HW) to be swapped. (See note.) n: The number of words to be swapped. Setting range is as follows: When specifying "n" directly : 1 to 16384 When specifying "n" indirectly: \$W0 to \$W32767 \$HW0 to \$HW8191 Note: Index can be set when using address to specify for S. Setting range is for \$W is between 0 and 32767 and for \$HW is between 0 to 8191..
Return Value	None
Example	SWAPL (\$W100, 3); Swap high order and low order of the word data which was taken from 3 long words from \$W100.

<Before executing SWAPL>

\$W100	1	2	3	4
\$W101	5	6	7	8
\$W102	9	0	1	2
\$W103	3	4	5	6
\$W104	7	8	9	0
\$W105	1	2	3	4

Sw ap high
order and low
order of long
word data

<After executing SWAPL>

\$W100	5	6	7	8
\$W101	1	2	3	4
\$W102	3	4	5	6
\$W103	9	0	1	2
\$W104	1	2	3	4
\$W105	7	8	9	0

WRITECF Saves the contents of a PT memory in a memory card (CF)

Applicable versions	System version 3 or higher
Format	WRITECF (Mem, Size, File, Dev)
Function	Saves the contents of PT memory in the specified file of Memory Card. The contents of the specified address will be written to the file in binary format. Mem: Top address of source data. (\$W, \$HW or \$SW) Index can be used. Setting range for \$W is between 0 and 32767, and for \$HW is between 0 and 8191. \$B, \$HB or \$SB can not be specified. Size: Data size to be saved in a Memory Card. (unit; word) Data size can be specified using long word directly, \$W or \$HW (uses 2 words) indirectly. Setting range for \$W is between 0 and 32767 and for \$HW is between 0 and 8191. If the set value has been exceeded the maximum, an access error will occur and an error message will appear File: Destination file name The file name can be specified using character string directly or using \$W or \$HW (uses 2 words) indirectly. If the specified file name for "F" already exists, the file name will be overwritten without showing a confirmation message. (Return value will be 0 (Completed normally)). Up to 43 alphanumerical characters ("0 to 9", "A to Z", "a to z", "\$", "_") including extension can be set for the file name. Dev: Always specify 0 since destination will be a Memory Card only.
Return Value	0: Completed normally -1: Failed to save data
Example	\$W100=WRITECF(\$W1000, 128, "CF_FILE.BIN", 0); \$W2000L=128; STRCPY(\$W2002, "CF_FILE.BIN"); \$W100=WRITECF(\$W1000, \$W2000, \$W2002, 0);

WRITECMEM Writes the data to the specified address

Applicable versions	System version 2 or higher
Format	WRITECMEM(D,S,n)
Function	Copy data of n bit/n channel from “S” to the address in the host specified in “D”. Maximum points of writing are indicated below.

Bit	126Bit
Word	126Channel

If the value outside of range is set for “n”, communication error or macro execution error occurs. Maximum points of writing depends on PLC type.

Return Value	None
Example	WRITECMEM([HOST1: DM0], \$W0,10); “Write the value \$W0 to \$W9 at the host named HOST1 in PLC.

WRITEHOSTB Writes bit data to the specified address

Applicable versions	System Version 6.2 or higher
Format	WRITEHOSTB(h, ch, addr, r, S, n)
Function	Copies n-bit data from (\$B/\$HB), S in the PT memory to the specified host, h. h: Host (host name/host number) ch: Host address type *1 addr: Host starting address r: Bits S: Source starting address (\$B0 to \$B32767, \$HB0 to \$HB8191) n: Number of elements to write (1 to 126) * 1: Refer to <i>Address Type Number</i> at the end of this chapter.
Return Value	Normal termination: 0x0000 Error: high order 8 bits (B8 to B15): MRES (main response cord) low order 8 bits (B0 to B7) : SRES (sub-response cord) *Refer to <i>5-2-7 Communications Errors and Countermeasures</i> in the <i>NS-Series Programming Manual</i> for MRES, and SRES.
Example	SerialA and Serial B are registered in the host: - Writes \$B10 to \$B19 to CIO1000.00 in the PLC connected to the host 1 (Serial port A). WRITEHOSTB(1, 100, 1000, 0, \$B10, 10); - Writes \$HB10 to \$HB19 to DM1000.05 in the PLC connected to the host name=[Serial B] (Serial port B). WRITEHOSTB([SerialB], 300, 1000, 5, \$HB10,10);

Reference

- ◆ If a host is deleted with CX-Designer, the subsequent hosts will be renumbered in order. The example below shows when HOST2 is deleted, the host numbers are renumbered and HOST3's number will be changed to 2.

Before Deleting Host		After Deleting Host	
Host Number	Host Name	Host Number	Host Name
1	SERIALA	1	SERIALA
2	HOST2	2	HOST3
3	HOST3	-	-

Please note that when a host number is given as argument h of WRITEHOSTB, it is necessary to specify the host number after change. With the above example, please change the macro as follows.

Macro Before Deleting Host	WRITEHOSTB(\$B10, <u>3</u> ,100,1000,0,10);	Change "3", the 1 st argument that specifies HOST3, to "2"
Macro After Deleting Host	WRITEHOSTB(\$B10, <u>2</u> ,100,1000,0,10);	

To check the host number, confirm the number at the left side of host name shown on the Comm.Setting dialog of CX-Designer.

- ◆ Also, note that if a host name is changed with CX-Designer when the host name is used as argument h of WRITEHOSTB, it is necessary to specify the host name after change.

For example, if a host name is changed from HOST_1 to HOST_A, change as follows.

Macro Before Changing Host Name	WRITEHOSTB(\$B10,[HOST_1],100,1000,0,10);	Change HOST_1 to HOST_A
Macro After Changing Host Name	WRITEHOSTB(\$B10,[HOST_A],100,1000,0,10);	

WRITEHOSTW Writes word data to the specified address

Applicable versions	System Version 6.2 or higher
Format	WRITEHOSTW(h, ch, addr, r, S, n)
Function	Copies n-word data starting (\$W/\$HW), S in the PT memory to the specified host, h. h: Host (host name/host number) ch: Host address type *1 addr: Host starting address S: Source starting address (\$W0 to \$W32767, \$HW0 to \$HW8191) n: Number of elements to write (1 to 126) * 1: Refer to <i>address type number</i> at the end of this chapter.
Return Value	Normal termination: 0x0000 Error: high order 8 bits (B8 to B15): MRES (main response cord) low order 8 bits (B0 to B7) : SRES (sub-response cord) *Refer to <i>5-2-7 Communications Errors and Countermeasures</i> in the <i>NS-Series Programming Manual</i> for MRES, and SRES.
Example	SerialA and Serial B are registered in the host: - Writes \$W10 to \$W19 to CIO1000 in the PLC connected to the host 1 (Serial port A). WRITEHOSTW(1, 100, 1000, \$W10, 10); - Writes \$HW10 to \$HW19 to DM1000 in the PLC connected to the host name=[Serial B] (Serial port B). WRITEHOSTW([SerialB], 300, 1000, \$HW10, 10);

Reference

- ◆ If a host is deleted with CX-Designer, the subsequent hosts will be renumbered in order. The example below shows when HOST2 is deleted, the host numbers are renumbered and HOST3's number will be changed to 2.

Before Deleting Host		After Deleting Host	
Host Number	Host Name	Host Number	Host Name
1	SERIALA	1	SERIALA
2	HOST2	2	HOST3
3	HOST3	-	-

Please note that when a host number is given as argument h of WRITEHOSTW, it is necessary to specify the host number after change. With the above example, please change the macro as follows.

Macro Before Deleting Host	WRITEHOSTW(\$B10, <u>3</u> ,100,1000,0,10);	Change "3", the 1 st argument that specifies HOST3, to "2"
Macro After Deleting Host	WRITEHOSTW(\$B10, <u>2</u> ,100,1000,0,10);	

To check the host number, confirm the number at the left side of host name shown on the Comm.Setting dialog of CX-Designer.

- ◆ Also, note that if a host name is changed with CX-Designer when the host name is used as argument h of WRITEHOSTW, it is necessary to specify the host name after change.

For example, if a host name is changed from HOST_1 to HOST_A, change as follows.

Macro Before Changing Host Name	WRITEHOSTW(\$B10,[HOST_1],100,1000,0,10);	Change HOST_1 to HOST_A
Macro After Changing Host Name	WRITEHOSTW(\$B10,[HOST_A],100,1000,0,10);	

Address Type Number

Address Type Name	Address Type Number	
	BCD	Binary
PT memory - \$B	0	0
PT memory - \$W	1	1
PT memory - \$SB	2	2
PT memory - \$SW	3	3
PT memory - \$HB	4	4
PT memory - \$HW	5	5
Data area (CIO)	100	64
Holding area (HR)	101	65
Auxiliary area (AR)	102	66
Link area (LR)	103	67
Work area (WR)	104	68
Timer (TIM) Only available for READHOSTW / WRITEHOSTW	200	C8
Counter (CNT) Only available for READHOSTW / WRITEHOSTW	201	C9
Data memory area (DM)	300	12C
Expansion data memory (EM)	301	12D
Expansion data memory 0 (EM0)	302	12E
Expansion data memory 1 (EM1)	303	12F
Expansion data memory 2 (EM2)	304	130
Expansion data memory 3 (EM3)	305	131
Expansion data memory 4 (EM4)	306	132
Expansion data memory 5 (EM5)	307	133
Expansion data memory 6 (EM6)	308	134
Expansion data memory 7 (EM7)	309	135
Expansion data memory 8 (EM8)	310	136
Expansion data memory 9 (EM9)	311	137
Expansion data memory A (EMA)	312	138
Expansion data memory B(EMB)	313	139
Expansion data memory C(EMC)	314	13A

The following macros can be used with NS-Runtime. The details of macros are described as below.

EXEC Application Startup

Applicable versions	System Version 6.6 or higher
Format	EXEC(S1, S2,S3)
Function	Executes the command specified with S1 and displays at S2 window title and in S3 window style Specify a startup file and a startup argument with S1. (Separate a startup file and a startup argument with a space) S2: Window title after a startup (" " displays the default title at startup) S3: (0=normal, 1=Minimize, 2=Maximize, 3=Hide) Use Unicode if you use symbols to specify strings with S1 and S2.
Return Value	0: Completed normally -1:Startup failed
Example	<pre>EXEC("CMD.EXE","ABC", 2); 'Execute CMD.EXE and maximize a window titled ABC.</pre>

STRCAT(W) String Concatenation

Applicable versions	System Version 6.6 or higher
Format	STRCAT(D, S) ...ASCII code STRCATW(D, S) ...Unicode
Function	Connects the string S to the string D.
Return Value	None
Example	<pre>STRCPY(\$W0, "ABC"); 'Set ABC to \$W0 to \$W1. STRCPY(\$W10, "DEF"); 'Set DEF to \$W10 to \$W11. STRCAT(\$W0, \$W10); 'Set ABCDEF to \$W0 to \$W3.</pre>

STRCMP(W), STRICMP(W) String Comparison

Applicable versions	System Version 6.6 or higher
Format	STRCMP(S1,S2) STRICMP(S1,S2) ...ASCII code STRCMPW(S1,S2) STRICMPW(S1,S2) ...Unicode
Function	Compares the string. STRCMP(W) Case sensitive STRICMP(W) Not case sensitive
Return Value	-1 : Disagree, S1<S2 0 : Agree, S1=S2 1 : Disagree, S1>S2
Example	STRCPY(\$W0, "ABC"); 'Set ABC to \$W0 to \$W1. \$W10= STRCMP(\$W0, "DEF"); 'Compare ABC and DEF. The result, -1, is stored in \$W10.

STRLEFT(W) Extracts the specified number of characters from leftmost characters of a string

Applicable versions	System Version 6.6 or higher
Format	STRLEFT(D,S,n), ...ASCII code STRLEFTW(D,S,n) ...Unicode
Function	Stores n characters from the left of the string S to D.
Return Value	None
Example	STRLEFT(\$W0,"ABCDEFGH",3); 'Extract 3 characters (ABC) from the leftmost string and store ABC in \$W0 to \$W1.

STRLEN(W) Gets string length

Applicable versions	System Version 6.6 or higher
Format	STRLEN(S) ···ASCII code STRLENW(S) ···Unicode
Function	Returns the length of the string S (The number of bytes of S).
Return Value	String Length
Example	STRCPY(\$W0, "ABC"); \$W10 = STRLEN(\$W0); 'Set 3 to '\$W10

STRLTRIM(W) Deletes the leftmost spaces of a string

Applicable versions	System Version 6.6 or higher
Format	STRLTRIM(D,S) ···ASCII code STRLTRIMW(D,S) ···Unicode
Function	Deletes the leftmost space of the string S to enter it to D.
Return Value	None
Example	STRLTRIM(\$W0, " ABC"); 'Store ABC excluding left spaces of a string in \$W0 to \$W1.

STRLWR(W) Converts a string to lower case

Applicable versions	System Version 6.6 or higher
Format	STRLWR(D, S) ···ASCII code STRLWRW(D, S) ···Unicode
Function	Converts upper cases of the string S to lower cases and enter them to D.
Return Value	None
Example	STRCPY(\$W0, "ABC"); STRLWR(\$W10, \$W0); 'Set abc to \$W10 to \$W11

STRMID(W) Extracts the specified number of characters from a specified character position of a string

Applicable versions	System Version 6.6 or higher
Format	STRMID(D,S,n1,n2) ...ASCII code STRMIDW(D,S,n1,n2) ...Unicode
Function	Extracts n2 characters from n1 of a string specified with S. Then store them in D. (n1: The head of a string is set to 1.)
Return Value	None
Example	STRMID(\$W0,"ABCDEFGH",2,3); 'Extracts 3 characters (BCD) from the 2nd of the string. Then store BCD in \$W0 to \$W1.

STRRIGHT(W) Extracts the specified number of characters from rightmost characters of a string

Applicable versions	System Version 6.6 or higher
Format	STRRIGHT(D,S,n) ...ASCII code STRRIGHTW(D,S,n) ...Unicode
Function	Extracts n characters from the rightmost characters of the string S. Then sets them in D.
Return Value	None
Example	STRRIGHT (\$W0,"ABCDEFGH",3); 'Extract 3 characters (EFG) from the rightmost characters of the string. Set EFG to \$W0 to \$W1.

STRRTRIM(W) Deletes the rightmost spaces of a string

Applicable versions	System Version 6.6 or higher
Format	STRRTRIM(D,S) ···ASCII code STRRTRIMW(D,S) ···Unicode
Function	Deletes the rightmost spaces of the string S to enter them to D.
Return Value	None
Example	STRRTRIM(\$W0, "ABC "); 'Set ABC to \$W0 to \$W1 excluding the rightmost spaces of the string.

STRTRIM(W) Deletes the spaces at both sides of a string

Applicable versions	System Version 6.6 or higher
Format	STRTRIM(D,S) ···ASCII code STRTRIMW(D,S) ···Unicode
Function	Extracts spaces at both sides of a string specified with S. Then stores them to D.
Return Value	None
Example	STRTRIM(\$W0, " ABC "); 'Set ABC in \$W0 to \$W1 excluding spaces at both ends.

STRUPR(W) Converts a string to upper case

Applicable versions	System Version 6.6 or higher
Format	STRUPR(D, S) ···ASCII code STRUPRW(D, S) ···Unicode
Function	Converts a string S from lower case to upper case. Then set it to D.
Return Value	None
Example	STRCPY(\$W0, "abc"); STRUPR(\$W10, \$W0); 'Set ABC to \$W10 to \$W11.

WINFIND Finds a window title

Applicable versions	System Version 6.6 or higher
Format	WINFIND(S1,S2)
Function	Searches whether a window specified with S1 has started or not. Set the following search conditions for each bit with S2. The 0 bit to 3rd bit: 0:Window title that completely matches with S1. 1:Window title that matches with the number of characters of S1. 2:Window title that matches with the number of characters of S1 (Except for a folder). The 4th bit: 0:The search ends when a target is found. 1:Searches for all the matched windows. Use Unicode if you use a symbol to specify a string with S1.
Return Value	The number of find results (0:None, 1 or more: Found)
Example	WINFIND("TEST",0); 'Search whether there is a window titled TEST.

WINMAX Maximizes a specified window

Applicable versions	System Version 6.6 or higher
Format	WINMAX(S1, S2)
Function	Maximizes a window specified with S1. Set the following search conditions for each bit with S2. The 0 bit to 3rd bit: 0:Window title that completely matches with S1. 1:Window title that matches with the number of characters of S1. 2:Window title that matches with the number of characters of S1 (Except for a folder). Use Unicode if you use a symbol to specify a string with S1.
Return Value	0 :Completed normally -1:No specified window
Example	WINMAX("TEST",0) ; 'Maximize a window titled TEST.

WINMIN Minimizes a specified window

Applicable versions	System Version 6.6 or higher
Format	WINMIN(S1, S2)
Function	Minimizes a window specified with S1. Set the following search conditions for each bit with S2. The 0 bit to 3rd bit: 0:Window title that completely matches with S1. 1:Window title that matches with the number of characters of S1. 2:Window title that matches with the number of characters of S1 (Except for a folder). Use Unicode if you use a symbol to specify a string with S1.
Return Value	0 :Completed normally -1:No specified window
Example	WINMIN("TEST",0) ; 'Minimize a window titled TEST.

WINNORMAL Restores a size of a specified window

Applicable versions	System Version 6.6 or higher
Format	WINNORMAL(S1, S2)
Function	Restores a size of a window specified with S1. Set the following search conditions for each bit with S2. The 0 bit to 3rd bit: 0:Window title that completely matches with S1. 1:Window title that matches with the number of characters of S1. 2:Window title that matches with the number of characters of S1 (Except for a folder). Use Unicode if you use a symbol to specify a string with S1.
Return Value	0 :Completed normally -1:No specified window
Example	WINNORMAL("TEST",0) ; 'Restore a size of a window titled TEST.

WINTERM Exits a specified window

Applicable versions	System Version 6.6 or higher
Format	WINTERM(S1, S2)
Function	Exits a window specified with S1. Set the following search conditions for each bit with S2. The 0 bit to 3rd bit: 0:Window title that completely matches with S1. 1:Window title that matches with the number of characters of S1. 2:Window title that matches with the number of characters of S1 (Except for a folder). The 4th bit: 0:Sends a WOM_CLOSE message to a specified window. 1: Sends a WM_DESTROY message to a specified window. Example: Microsoft Word When the 4th bit is 0, Displays a message saying Do you want to save the document 1? After confirming, Microsoft-Word ends. When the forth bit is 1, Exits a window without displaying a confirmation message even when there is a change. Use Unicode if you use a symbol to specify a string with S1.
Return Value	0:Completed normally -1:No specified window
Example	WINTERM("TEST",0); 'Exit a window titled TEST.

WINTOP **Brings a specified window to the front**

Applicable versions	System Version 6.6 or higher
Format	WINTOP(S1, S2)
Function	Brings a window specified with S1 to the front. Set the following search conditions for each bit with S2. The 0 bit to 3rd bit: 0:Window title that completely matches with S1. 1:Window title that matches with the number of characters of S1. 2:Window title that matches with the number of characters of S1 (Except for a folder). Use Unicode if you use a symbol to specify a string with S1.
Return Value	0: Completed normally -1:No specified window
Example	WINTOP("TEST",0) ; 'Bring a window titled TEST to the front.

Section3 Error Message List

This section describes error message which is displayed in error list box when macro is added to the project, the screen and the functional objects.

3-1	Error Message List	53
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Error Message List

Error messages are displayed in the error list box after checking the error as shown below.

Error Message	Details	Example
Format error	The program contains description besides variable name, function name, or programming terms that cannot be recognized. Check whether the input function is correct.	\$W0=ABC+100;
Variable name error	Variable name is incorrect	\$B0:3=1;
(is missing	The ((left parentheses symbol) is missing from a function or sentence	IF\$W0==1)
No. of ()does not agree	The number of () (parentheses) in the program does not agree	IF(\$W0=1)!(\$W1=0
Position of , is incorrect	The position of the , (comma) is incorrect	IF(\$W0==1),(\$W1==0)
Function argument error	The program contains an incorrect function argument, such as word memory being set in a position that permits bit memory only. Refer to "Section 2 Explanation for the function "-Correspondence Table of Function and Argument" and check the argument.	\$W0=BCD(\$B0);
= Command error	The program contains an incorrect substitution statement, such as 3=10, \$B0=3	\$W0="ABCDE"
End of program is incomplete	The program that was input is incomplete	\$W10=10+;
If sentence error	The program contains an incorrect IF,ELSE or ENDIF is statement	IF(\$W0==1)!(\$W1==0) \$W10=1; ELSE \$W10=10;
,or; is missing	The number of ,(comma) that divides the argument is insufficient. The program is not divided by a ;(semicolon).	\$W10=1
FOR Statement is mismatch	FOR statement is not closed by NEXT	FOR(3) \$W0=\$W0+1;
	BREAK or CONTINUE is placed outside of FOR loops.	FOR(3) \$W0=\$W0+1; NEXT; BREAK;
Nest of FOR statement is exceeded the max. (Single loop only)	FOR is nested too deep. Nesting of loops is not supported.	FOR(3) \$W0=\$W0+1; FOR(5) \$W10=\$ W 10+10; NEXT; NEXT;

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